

TASK1: To Register Handlers, Configure Signal Actions, Handle Interrupts, Manage Signal Context, Verify Stability, Test Signal Events

```
jhansi_priya@Jhansi-Priya: ~/ $
```

```
jhansi_priya@Jhansi-Priya:~$ mkdir Skill10  
jhansi_priya@Jhansi-Priya:~$ cd Skill10  
jhansi_priya@Jhansi-Priya:~/Skill10$ nano task1.c  
jhansi_priya@Jhansi-Priya:~/Skill10$ gcc task1.c -o task1  
jhansi_priya@Jhansi-Priya:~/Skill10$ ./task1  
Process started.  
PID: 621  
Press Ctrl+C to send SIGINT.  
Program is running...  
Program is running...  
Program is running...  
^C  
Signal received: 2  
Signal handled successfully!  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
^C  
Signal received: 2  
Signal handled successfully!  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Program is running...  
Terminated  
jb@nsdya@Jhansi-Priya:~/Skill10$
```

```
jhansi_priya@Jhansi-Priya: ~/  × + ▾  
jhansi_priya@Jhansi-Priya:~$ cd Skill10  
jhansi_priya@Jhansi-Priya:~/Skill10$ kill -TERM 521  
jhansi_priya@Jhansi-Priya:~/Skill10$ |
```

TASK2: To Capture SIGINT, Forward Signals, Protect Shell Process, Terminate Foreground Jobs, Update Job States, Test Interrupt Handling

```
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void sigint_handler(int sig)
{
    printf("\nSIGINT received!\n");
    printf("The program received Ctrl+C.\n");
}

int main()
{
    signal(SIGINT, sigint_handler);

    printf("Program PID: %d\n", getpid());
    printf("Press Ctrl+C to test SIGINT.\n");

    while (1)
    {
        printf("Foreground job is running...\n");
        sleep(3);
    }

    return 0;
}
```

```
jhansi_priya@Jhansi-Priya:~/Skill10$ nano task2.c
jhansi_priya@Jhansi-Priya:~/Skill10$ gcc task2.c -o task2
jhansi_priya@Jhansi-Priya:~/Skill10$ ./task2
Program PID: 566
Press Ctrl+C to test SIGINT.
Foreground job is running...
Foreground job is running...
^C
SIGINT received!
The program received Ctrl+C.
Foreground job is running...
Foreground job is running...
Foreground job is running...
^C
SIGINT received!
The program received Ctrl+C.
Foreground job is running...
^Z
[1]+  Stopped                  ./task2
jhansi_priya@Jhansi-Priya:~/Skill10$ jobs
[1]+  Stopped                  ./task2
jhansi_priya@Jhansi-Priya:~/Skill10$ bg %1
[1]+ ./task2 &
Foreground job is running...
jhansi_priya@Jhansi-Priya:~/Skill10$ j
Foreground job is running..
..
obs
[1]+  Running                  ./task2 &
jhansi_priya@Jhansi-Priya:~/Skill10$
Foreground job is running...
```

```

jhansi_priya@Jhansi-Priya:~/Skill10$ Foreground job is running..
Foreground job is running...
fForeground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
^C
jhansi_priya@Jhansi-Priya:~/Skill10$ Foreground job is running..
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
Foreground job is running...
^C
[1]+  Terminated                  ./task2
jhansi_priya@Jhansi-Priya:~/Skill10$ |

```

TASK3: To Capture SIGTSTP, Suspend Processes, Update Job Table, Preserve State, Resume Later, Test Suspension Workflow

```

GNU nano 7.2 task3.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void sigtstp_handler(int sig)
{
    printf("\nSIGTSTP received!\n");
    printf("Process received Ctrl+Z.\n");
}

int main()
{
    signal(SIGTSTP, sigtstp_handler);

    printf("Process PID: %d\n", getpid());
    printf("Press Ctrl+Z to send SIGTSTP.\n");

    while (1)
    {
        printf("Process is running...\n");
        sleep(2);
    }

    return 0;
}

```

```

jhansi_priya@Jhansi-Priya:~/Skill10$ nano task3.c
jhansi_priya@Jhansi-Priya:~/Skill10$ gcc task3.c -o task3
jhansi_priya@Jhansi-Priya:~/Skill10$ ./task3
Process PID: 706
Press Ctrl+Z to send SIGTSTP.
Process is running...
Process is running...
Process is running...
^Z
SIGTSTP received!
Process received Ctrl+Z.
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
^Z
SIGTSTP received!
Process received Ctrl+Z.
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
Process is running...
Terminated

```

TASK4: To Create Process Groups, Assign Group IDs, Transfer Terminal Control, Restore Ownership, Coordinate Signals, Validate Behavior

```

GNU nano 7.2 task4.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    pid_t pid = getpid();
    pid_t ppid = getppid();
    pid_t pgid = getpgrp();
    pid_t sid = getsid(0);

    printf("Process ID (PID): %d\n", pid);
    printf("Parent PID: %d\n", ppid);
    printf("Process Group ID (PGID): %d\n", pgid);
    printf("Session ID (SID): %d\n", sid);

    printf("\nProcess is running...\n");

    while (1)
    {
        sleep(2);
    }

    return 0;
}

```

```
jhansi_priya@Jhansi-Priya: ~/ x + v
jhansi_priya@Jhansi-Priya:~/Skill10$ nano task4.c
jhansi_priya@Jhansi-Priya:~/Skill10$ gcc task4.c -o task4
jhansi_priya@Jhansi-Priya:~/Skill10$ ./task4
Process ID (PID): 1180
Parent PID: 388
Process Group ID (PGID): 1180
Session ID (SID): 388

Process is running...
^C
jhansi_priya@Jhansi-Priya:~/Skill10$ ps -o pid,ppid,pgid,sid,tpgid,stat,cmd
  PID    PPID    PGID    SID    TPGID  STAT  CMD
   388     387     388     388     1181   Ss    -bash
  1160     388     1160     388     1181   S     sleep 100
  1181     388     1181     388     1181  R+    ps -o pid,ppid,pgid
jhansi_priya@Jhansi-Priya:~/Skill10$ kill -TERM -1180
-bash: kill: (-1180) - No such process
[1]+  Done                  sleep 100
jhansi_priya@Jhansi-Priya:~/Skill10$ kill -TERM -1160
-bash: kill: (-1160) - No such process
jhansi_priya@Jhansi-Priya:~/Skill10$ ./task4 &
[1] 1191
jhansi_priya@Jhansi-Priya:~/Skill10$ Process ID (PID): 1191
Parent PID: 388
Process Group ID (PGID): 1191
Session ID (SID): 388

Process is running...
jobs -l
[1]+  1191 Running          ./task4 &
jhansi_priya@Jhansi-Priya:~/Skill10$ kill -STOP 1191
jhansi_priya@Jhansi-Priya:~/Skill10$ jobs
[1]+  Stopped              ./task4
jhansi_priya@Jhansi-Priya:~/Skill10$ kill -CONT 1191
jhansi_priya@Jhansi-Priya:~/Skill10$ jobs
[1]+  Running              ./task4 &
jhansi_priya@Jhansi-Priya:~/Skill10$ |
```


